Lab 2

KTH ROYAL INSTITUTE
OF TECHNOLOGY

Data processing and modeling

General Approach for Data Processing

- Data Cleaning and Imputation
- Data Transformation
- Data Reductoin

Goals of Data Processing

Turn raw data into model-ready data

- Remove errors, missing values, and noise
- Make features comparable and numerically stable
- Reduce redundancy and unnecessary complexity

Disscussion

- Ways to deal with missing or incorrect values?
- How to choose appropriate methods to deal with missing or incorrect values

Data Processing

- To file lab2_processing.ipynb

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Data Processing

DYO:

Play around with lab2_processing, ask questions

Break:

From 11:00 -11:15

General Approach for Modeling

- Data Preparation
- Training Model
- Creating a Pipeline
- Evaluating
- Checking Generalization (Overfitting vs Underfitting)

Data Preparation

- Simple Split
  - Suppose we have enough data…
  - We’ve selected the right training data…
  - We’ve picked the features we want the model to learn from…
  - But how do we know our model will generalize well?

- K-Fold Validation
  - Split your data randomly into k equal size parts.
  - For each part, hold one back as a test set and train on the $k - 1$ remaining parts, evaluate on the part you held back.
  - Report the average evaluation.

K fold cross validation

- If k = 5, then you have 5 parts T1, . . . , T5 you would run 5 training runs
- Train on T1,T2,T3,T4 evaluate on T5.
- Train on T1,T2,T3,T5 evaluate on T4.
- Train on T1,T2,T4,T5 evaluate on T3.
- Train on T1,T3,T4,T5 evaluate on T2.
- Train on T2,T3,T4,T5 evaluate on T1.
- Good values of k are 5 or 10. Obviously the larger k is the more time it takes to run the experiments.

After K fold

- Cross validation only returns a value that is a prediction of **how well the model will do** on more data.
- k-fold cross validation without ensemble methods only tells you **which model is better.** It does not give you a trained model.
- Once you have chosen the parameters and model, **repeat the process to get an actual trained model.**

Training Model

- To file lab2_modeling.ipynb

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Terminology

- Type of **classification outputs**:
- True positive (m11): Example of class 1 predicted as class 1.
- False positive (m01): Example of class 0 predicted as class 1. Type 1 error.
- True negative (m00): Example of class 0 predicted as class 0.
- False negative (m10): Example of class 1 predicted as class 0. Type II error.
- Total number of instances: m = m00 + m01 + m10 + m11
- Error rate: (m01 + m10) / m
- If the classes are imbalanced (e.g. 10% from class 1, 90% from class 0), one can achieve low error (e.g. 10%) by classifying everything as coming from class 0!

Evaluating: Confusion Matrix

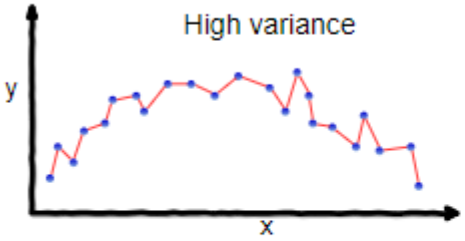
- [TrueNegative FalsePositive

  FalseNegative TruePositive]
- Do this for every class – multi class classification problems
- Format not consistent, be careful

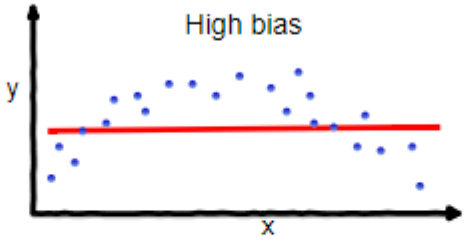
		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

Overfitting vs Underfitting

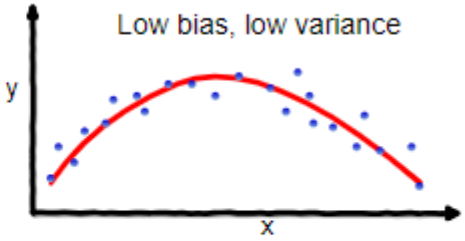
- Over-fitting: Training accuracy is higher (error is lower) than testing accuracy
  - Model explains the training data very well, but poor generalization
- Under-fitting: Both accuracies are unacceptable low



overfitting



underfitting



Good balance

Modeling

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Play around with lab2_modeling, ask questions